

JOURNAL CLUB — SECTION 5

Instruction-Tuning, Fine-Tuning, and Safety in Clinical LLMs

Three studies on adapting large language models for medicine

Stanley Liang

Three Studies, One Question

How should we adapt a large language model for medical use?

01 Instruction-Tuning for Biomedical Tasks

Teaching a model to follow clinical instructions — BioInstruct

02 The Fine-Tuning Fallacy

Why full-weight fine-tuning can quietly degrade clinical reasoning

03 Engineering Safety in Summarization

Driving hallucination and omission below the human baseline

TOPIC 01

Instruction-Tuning for Biomedical Tasks

Can natural-language instructions replace expensive, overfitting-prone fine-tuning?

Why Instruction Tuning Outperforms Traditional Fine-Tuning

TRADITIONAL FINE-TUNING	INSTRUCTION TUNING
Requires massive, high-quality task-specific datasets	Uses natural language as a universal interface
Produces highly specialized, narrow outputs	Adapts to unseen, novel clinical tasks
Prone to catastrophic overfitting on low-quality clinical data	Requires significantly less task-specific data
Result: a rigid, domain-specific oracle	Result: a flexible assistant that generalizes

Building BioInstruct: A Self-Instruct Pipeline

80 human-curated seed tasks ~3 hours
of manual effort



GPT-4 generation engine prompted with
3 random seeds / iteration



Heuristic filtering dedup, invalid lengths,
image-based

25,005

diverse instruction triplets generated
with zero additional human labeling

33.8%

Information Extraction

33.5%

Text Generation

22.8%

Question Answering

0.83

Krippendorff's alpha (3
raters)

Parameter-Efficient Training with LoRA

Frozen pretrained weights (LLaMA 1 & 2) plus small trainable rank-decomposition matrices — adapting the model with extreme hardware efficiency while preventing catastrophic forgetting of foundational medical knowledge.

TASK CATEGORY	CLINICAL GOAL	EVALUATION METRICS
Question Answering & NLI	Medical knowledge retrieval	Accuracy
Information Extraction	Clinical data identification	Precision, Recall, F1, Conditional Accuracy
Text Generation	Applied clinical skills (summarization / QA)	GPT-4 human-aligned eval, BERTScore, Concept_F1, Bleurt

Results: Gains Across QA, Extraction, and Generation

+17.3%

Average zero-shot accuracy —
Question Answering (MedQA,
MedMCQA, PubMedQA)

+5.7%

Overall F1 gain — Information
Extraction, driven by a recall jump
from 71.95 to 84.43

+96%

Overall GPT-4-aligned
performance leap — Text
Generation (coherence,
completeness)

BioInstruct Outperforms Models Trained on Millions of Documents

MODEL	TRAINING DATA	QA TASK DOMINANCE
Asclepius (7B)	Synthetic medical literature	Outperformed by BioInstruct
ChatDoctor	Dialogue datasets	Outperformed by BioInstruct
MedAlpaca	Alpaca medical tuning	Outperformed by BioInstruct
PMC-LLaMA (7B)	4.8M academic papers & 30k textbooks	Outperformed by BioInstruct
BioInstruct-LLaMA (7B)	Just 25,005 auto-generated instructions	Top performer in 4 of 4 tasks

High-quality, tailored instruction tuning proves far more efficient than brute-force domain pre-training.

The Data Scaling Law: Most Gains Arrive by 5,000 Examples

Scaling behavior

- Steepest performance gains occur almost immediately, peaking near 5,000 instructions
- Generation quality scales roughly linearly with data volume
- QA and extraction hit generalization hurdles at 10k before stabilizing near 25k

Cross-task synergy

Training on QA → **+2.04% IE, +2.15% Generation**

Training on Generation → **-0.94% QA, -3.74% IE**

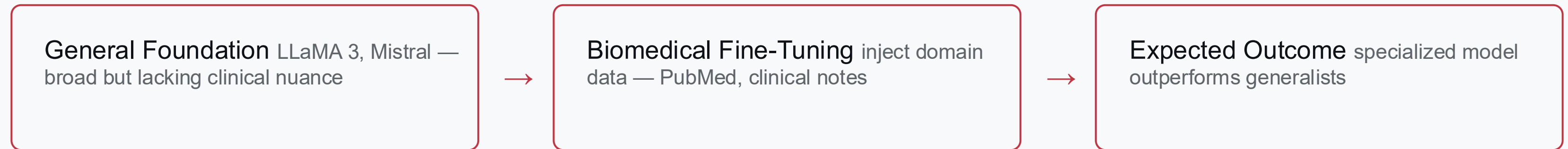
Format beats volume — but which tasks you mix in still matters.

TOPIC 02

The Fine-Tuning Fallacy

Why general-purpose models are outperforming specialized medical models in the clinic

The Assumption: Medical Tasks Require Medical Models



Healthcare leaders assume fine-tuning is mandatory for safety and accuracy. Is it?

Benchmark Contamination Masks True Performance

The Illusion

- Biomedical models achieve state-of-the-art scores on USMLE-style benchmarks
- These exams have been online for years
- Models may be memorizing test data, not reasoning

The Reality

- Real clinical practice is chaotic and fragmented
- Requires extracting sparse data, assigning ICD-10 codes, true clinical reasoning
- Poorly represented by standardized multiple-choice exams

Testing 14 Models on 8 Unseen Clinical Tasks

Benchmarks chosen explicitly outside the fine-tuning datasets, to prevent data leakage.

Pillar 1: Medical Knowledge

NEJM & JAMA case challenges —
diagnosing complex, real-world clinical
vignettes

Pillar 2: Extraction & Coding

MeDiSumQA, MeDiSumCode,
ProblemSummary — assigning accurate ICD-
10 codes from notes

Pillar 3: Clinical Reasoning

MedNLI, LongHealth, MeQSum — logical
relationships and identifying missing
information

Fine-Tuning Degrades Clinical Reasoning

64.3%

Llama-3-8B (Generalist) NEJM case-challenge accuracy



30.0%

OpenBioLLM-8B (Medical) same benchmark, after fine-tuning

On complex, unseen diagnostic cases, the fine-tuned biomedical model performed at less than half the accuracy of its general-purpose foundation. Mistral-7B (69.6%) → BioMistral-7B (58.0%) shows the same penalty.

CLINICAL TASK	GENERAL-PURPOSE LLM	BIOMEDICAL FINE-TUNED
Clinical Vignettes (JAMA, 70B)	Tie (65.0%)	Tie (66.4%)
Discharge Summarization	Wins	Loses
ICD-10 Coding	Wins	Loses
Consumer Health Queries	Wins	Loses

The Mechanism: Catastrophic Forgetting

Generalist models are trained on up to **15 trillion tokens** — nearly the totality of freely available biomedical text (PubMed, MIMIC, and more) is likely already inside the baseline model.

Fine-tuning on that same public data forces the model to redundantly "re-learn" facts it already knows — while overwriting the broad reasoning weights that made it a good generalist in the first place.

Why 70B models fared better

Larger parameter counts mean smaller *relative* weight updates during fine-tuning — 70B models absorbed new medical data with far less disruption to existing structure than 8B models, which suffered severe catastrophic forgetting from the same update.

Fine-Tuning Increases Hallucination Risk

LongHealth Task 3 tests whether a model can correctly state that information is missing from a patient record, rather than inventing an answer.

Ability to identify missing information

OpenBioLLM-70B (Medical)

Lower — higher hallucination risk

Llama-3-70B (Generalist)

Higher — lower hallucination risk

Medical fine-tuning increased the fabrication of facts — the opposite of what it was meant to do.

The Alternative: Retrieval-Augmented Generation

DIMENSION	FINE-TUNING	RAG
Reasoning & logic	Degraded via catastrophic forgetting	Fully preserved from generalist base
Knowledge source	Baked into weights, hard to update	External, instantly updatable
Hallucination risk	Increased — model invents missing info	Minimized via exact document retrieval
Best use case	Narrow formatting or tone adaptation	Clinical support, QA, summarization

Do not compromise a model's general reasoning to teach it medical facts.

TOPIC 03

Engineering Safety in Clinical Summarization

Driving hallucination and omission in LLM-generated notes below the human baseline

The Dual Threat: Hallucinations and Omissions

1.47%

Hallucinations

of generated sentences invent information not in the input transcript. 44% are classified Major. Danger zone: the "Plan" section.

3.45%

Omissions

of generated sentences drop relevant clinical information. 16.7% are classified Major. Danger zone: the "Current Issues" section.

The human baseline: clinician-written notes average at least **1 error and 4 omissions per note** — the bar AI must clear before it can be trusted.

A Taxonomy of Hallucinations

TYPE	SHARE	DEFINITION
Fabrication	43%	Inventing facts outright
Negation	30%	Reversing a stated clinical fact
Contextual	17%	Mixing in unrelated topics
Causality	10%	Speculating unstated causes

Input transcript: *"Patient says I have no chest pain."*



LLM note: "Patient presents with chest pain."

Negation flips are the highest clinical threat — they directly contradict the record.

Mapping the Danger Zones

Where major errors concentrate

Major hallucinations → **"Plan" section (21%)**

Major omissions → **"Current Issues" (55%)** and past medical / family / social history (35%)

Methodology — CREOLA framework

18 controlled experiments; **12,999 clinical-note sentences** manually reviewed by clinicians and scored by likelihood × consequence — the largest evaluation of LLM clinical-note generation to date. Goal: push every output into the negligible-risk zone.

Iterative Prompt Engineering Toward Structured Output

STAGE	METHOD	RESULT
Basic instruction (Exp 1)	Simple command to summarize	High baseline error rate
Structured SOAP + "unknown" constraint (Exp 3–8)	Impose SOAP template; require "unknown" if information is missing	Major hallucinations –75%, major omissions –58%
JSON function calling (Exp 6–11)	Strict schema with explicit negative constraints	Major omissions eliminated entirely (61 → 0)

Achieving Sub-Human Error Rates

METRIC	TYPICAL HUMAN NOTE	BASELINE LLM (EXP 1)	CREOLA-REFINED LLM (EXP 11)
Errors / note	~1	High variability, unsafe	2 major hallucinations across 25 notes
Omissions / note	~4	Frequent, uncontrolled	0 major omissions across 25 notes

Hallucinations are inevitable. Clinical harm is preventable — through structure, not model choice.

Three Studies, One Principle

- Instruction-tuning on well-designed, task-specific examples can outperform brute-force pretraining — teach the model *how* to respond, not just *what* to know.
- Full-weight fine-tuning risks erasing the general reasoning a model already has. Pairing a strong generalist with retrieval preserves it.
- Whether adapting or querying a model, structure and constraints — not scale — are what keep clinical outputs safe.

Preserve a model's reasoning. Control what it's allowed to say.